User Manual: Setting up MRTG Service

This guide will walk you through the process of setting up the MRTG (Multi Router Traffic Grapher) service on your system. MRTG is a tool used to monitor and graph network traffic.

1. System Update and Upgrade

First, we need to make sure our system is up-to-date. Open your terminal and run the following commands:

```
sudo apt update
sudo apt upgrade -y
```

2. Install Required Packages

We need to install `net-tools`, `snmpd`, and `snmp`. `snmpd` is the SNMP agent that will provide the data to MRTG.

```
sudo apt install net-tools
sudo apt install snmpd snmp
```

3. Configure SNMP

Next, we need to configure the SNMP agent.

3.1. Backup the original configuration file

It's always a good practice to backup the original configuration file before making any changes.

```
cd /etc/snmp
sudo cp snmpd.conf snmpd.conf.orig
```

3.2. Edit the configuration file

Open the `snmpd.conf` file in a text editor. The log file uses `vi`, but you can use any editor you are comfortable with (like `nano`).

```
sudo vi /etc/snmp/snmpd.conf
```

You need to make the following changes to the file:

- Find the line `#rocommunity public default -V systemonly` and comment it out (add a `#` at the beginning if it's not already there).
- Add a new line `rocommunity public default`.
- Add a new line `disk /` to enable disk monitoring for the root partition.
- Find the line `#includeAllDisks 10%` and uncomment it (remove the `#`).

The `diff` in the log file shows the changes that were made. You can use it as a reference.

3.3. Start the SNMP service

Now, we can start the `snmpd` service and check its status.

```
sudo systemctl restart snmpd
sudo systemctl status snmpd
```

3.4. Test SNMP

Let's test if SNMP is working correctly. The following command should return the system information.

```
snmpwalk -v2c -c public localhost .1.3.6.1.2.1.1.1.0
```

4. Install and Configure MRTG

Now we will install and configure MRTG.

4.1. Install Apache and MRTG

MRTG uses a web server to display the graphs. We will use Apache.

```
sudo apt install apache2
sudo apt install mrtg
```

4.2. Create the MRTG configuration file

The `cfgmaker` command creates the MRTG configuration file. We will also set the `WorkDir` to `/var/www/html/mrtg`.

```
sudo cfgmaker public@localhost | sudo tee /etc/mrtg/mrtg.cfg > /dev/null
sudo sed -i '1iWorkDir: /var/www/html/mrtg' /etc/mrtg/mrtg.cfg
```

4.3. Add custom monitoring targets

We will add custom targets to monitor CPU, RAM, and disk usage.

4.3.1. Find the disk index Before adding the disk target, you need to find the correct index for the root partition (`/`). Use the following command:

```
snmpwalk -v2c -c public localhost .1.3.6.1.4.1.2021.9.1.2
```

Look for the line that ends with `STRING: "/"`. The number at the end of the OID on that line is the index for the root partition. For example, if you see `iso.3.6.1.4.1.2021.9.1.2.1 = STRING: "/"`, the index is `1`.

4.3.2. Append the targets Append the following content to the `/etc/mrtg/mrtg.cfg` file. **Remember to replace `<disk_index>` with the index you found in the previous step.**

```
Target[cpu]: .1.3.6.1.4.1.2021.11.50.0&.1.3.6.1.4.1.2021.11.52.0:public@localhost
Title[cpu]: CPU Usage (%)
PageTop[cpu]: <h1>CPU Usage</h1>
MaxBytes[cpu]: 100
YLegend[cpu]: CPU Usage (%)
ShortLegend[cpu]: %
Legend1[cpu]: User CPU
Legend2[cpu]: System CPU
Options[cpu]: gauge, nopercent

Target[ram]: .1.3.6.1.4.1.2021.4.6.0&.1.3.6.1.4.1.2021.4.15.0:public@localhost
Title[ram]: Memory Usage (KB)
PageTop[ram]: <h1>Memory Usage</h1>
MaxBytes[ram]: 1000000000
YLegend[ram]: Memory (KB)
ShortLegend[ram]: KB
Legend1[ram]: Available RAM
Legend2[ram]: Cached RAM
Options[ram]: gauge, nopercent

Target[disk]: .1.3.6.1.4.1.2021.9.1.9.<disk_index>&.1.3.6.1.4.1.2021.9.1.9.<disk_index>:publ
Title[disk]: Disk Usage (Root Partition - /)
PageTop[disk]: <h1>Disk Usage (Root Partition - /)</h1>
MaxBytes[disk]: 100
YLegend[disk]: Disk Usage (%)
ShortLegend[disk]: %
Legend1[disk]: Disk Usage
Legend2[disk]: Disk Usage (same)
LegendI[disk]: Used
LegendO[disk]:
Options[disk]: gauge, nopercent, noo
```

4.4. Run MRTG

Run MRTG to generate the initial graphs. You need to run it three times.

```
sudo env LANG=C /usr/bin/mrtg /etc/mrtg/mrtg.cfg
sudo env LANG=C /usr/bin/mrtg /etc/mrtg/mrtg.cfg
sudo env LANG=C /usr/bin/mrtg /etc/mrtg/mrtg.cfg
```

4.5. Create the index page

The `indexmaker` command creates an index page for the MRTG graphs.

```
sudo indexmaker --output=/var/www/html/mrtg/index.html /etc/mrtg/mrtg.cfg
```

You can view the index page by opening `http://<your_server_ip>/mrtg/` in your web browser.

5. Automate MRTG Data Collection

Finally, we will set up a cron job to run MRTG every 5 minutes to keep the graphs updated.

```
echo "*/5 * * * * root env LANG=C /usr/bin/mrtg /etc/mrtg/mrtg.cfg" | sudo tee /etc/cron.d/m
```

Congratulations! You have successfully set up the MRTG service.